Categorical Variables, Nonlinear Terms & Interaction Terms

STAT 223 – Applied Analytics

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Categorical Variables in Regression Models

All the regression models we've discussed so far have only dealt with numerical predictor variables.

But often, there are variables that would be important to predicting a response variable that can't be expressed numerically.

Categorical Predictor Variable

A **categorical predictor variable** has qualitative values representing one of a finite number of levels.

For example, consider height on sex or the vote share a candidate gets on race, sex or education status.

Categorical Variables in Regression Models

Dummy Indicator Variable

A **dummy indicator variable** is a binary variable that is 1 for a specified category of a categorical predictor and 0 for all other categories.

If a categorical predictor variable has k levels, how many dummy variables do we need to represent it?

Categorical Variables in Regression Models

If you have k levels for your variable, it can be represented by $k - 1$ dummy variables.

Why? Because when all the $k - 1$ dummy variables are 0, that represents the k^{th} level.

Reference Level

The category represented by the omitted level is known as the **reference level** of the categorical variable.

$$Y = \beta_0 + \beta_1 c_1 + \beta_2 c_2 + \beta_3 c_3 + \varepsilon$$

What are the interpretation of β_1, β_2 and β_3?

Regression with Nonlinear Terms

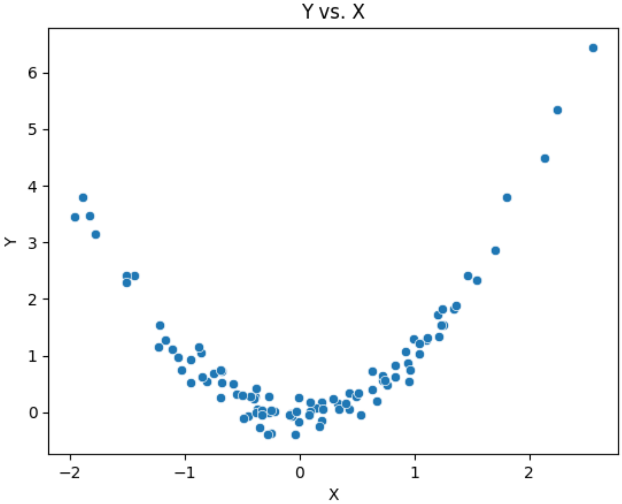
A simple linear regression uses a straight-line relationship between a response variable Y and a predictor variable X. But the data may have a nonlinear relationship.

For instance, age on a player's performance, or the month on the average daily temperature.

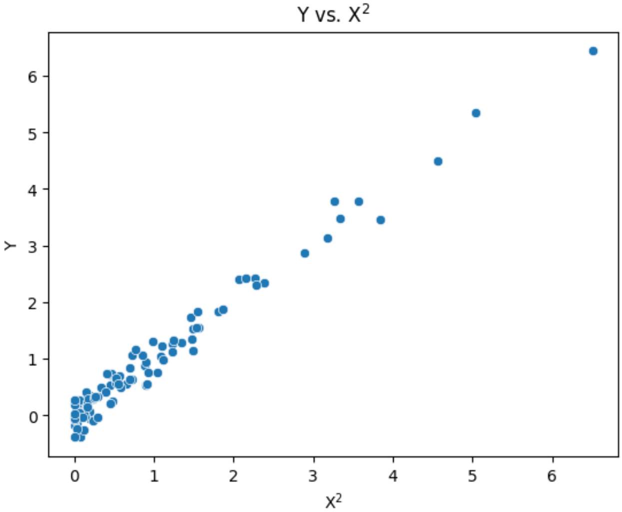
To do this, we can add powers of X_i to the regression equation.

$$\begin{aligned} \text{Quadratic Regression:} \quad & Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon \\ \text{Cubic Regression:} \quad & Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon \\ & \vdots \end{aligned}$$

Regression with Nonlinear Terms



Regression with Nonlinear Terms



Interaction in Regression Models

What if two predictor variables have an impact on each other in a way that affects the response variable but can't be explained on terms of the variable on their own?

For example, suppose that you are modeling the risk of a heart attack, and your explanatory variables are age and cholesterol. Not only does the higher cholesterol and higher age increase your risk for a heart attack, the impact of high cholesterol is larger as a person ages.

Reference Level

An **interaction term** is the product of two predictor variables in a multiple regression model and can be taken as another predictor variable.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

What is the interpretation of β_3?

Interaction in Regression Models

You can also have interaction in

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 c_1 + \beta_3 c_2 + \beta_4 c_3 + \beta_5 X_1 c_1 + \beta_6 X_1 c_2 + \beta_7 X_1 c_3 + \varepsilon$$

What are the interpretations of β_5, β_6 and β_7?